

Use-Case Flow Specification

Use Case: UC-04 — Claim Found Item

Primary Actor: Owner (claimant) Supporting Actors: System (matching/verification engine), Finder

Preconditions

- The Owner has an active account and is authenticated in the system.
- A Found Item post exists with status “Active” and has at least one candidate match generated against the Owner's Lost Item report (UC-03).
- The Finder has set at least one private security question on the Found Item post (precondition for UC-05).
- The Owner does not already have a verified claim on the same Found Item post.

Postconditions

Success (guaranteed):

- A claim record exists with status “Verified,” a unique claim ID, and a creation timestamp.
- The claim record is linked to both the Lost Item report and the Found Item post.
- The Found Item listing is flagged “Claim Verified – Pending Handover” so other applicants see its status.
- Both parties' contact details are exchanged (UC-06) and a confirmation notification/email is sent to each.

Failure (minimal guarantee):

- No claim record is created; the Found Item post's status and any other applicants' pending claims are left unchanged.
- The Owner is shown the specific reason the claim did not succeed and may retry.

Business Rules

- BR-1: A Finder may set at most one active security-question set per Found Item post.
- BR-2: An Owner may have multiple open claim attempts across different Found Item posts, but only one verified claim per Lost Item report.
- BR-3: Only Found Item posts with status “Active” are eligible for a claim; posts already under a verified claim are excluded.
- BR-4: Security-question answers are stored hashed and are never redisplayed to the Finder after initial entry.

Main Success Scenario

1. The Owner selects “This is my item” on a candidate match surfaced from UC-03, on a Found Item post marked “Active.”
2. The system verifies the Owner is authenticated; an already-authenticated session proceeds directly to step 3.
3. The system checks eligibility: the Found Item post is still “Active” and not already under a verified claim (BR-2, BR-3).

4. The system displays the security question(s) set by the Finder for that post (UC-05: Verify Ownership).
5. The Owner submits answers to the security question(s).
6. The system validates the submitted answers against the Finder's stored, hashed answers (BR-4).
7. Validation succeeds; the system creates a claim record with status "Verified," a timestamp, and links to both posts.
8. The system flags the Found Item listing as "Claim Verified – Pending Handover."
9. The system reveals the Finder's contact details to the Owner and the Owner's contact details to the Finder (UC-06).
10. The system sends a confirmation notification/email to both parties summarizing next steps for handover.
11. Use case ends successfully.

Alternate Flows

2a. Owner Is Not Authenticated

- 2a1. The system redirects the Owner to the login/registration screen.
- 2a2. On successful authentication, the system returns the Owner to the match screen and resumes at step 3.
- 2a3. If the Owner cancels login, the use case ends without a claim being submitted.

3a. Found Item No Longer Eligible

- 3a1. The system detects the post's status has changed (e.g., already claimed or removed) since the Owner began the process.
- 3a2. The system informs the Owner and suggests other similar candidate matches (UC-03).
- 3a3. Use case ends without a claim being submitted.

6a. Verification Fails

- 6a1. The system displays an "Answers did not match" message and does not reveal any contact details.
- 6a2. The Owner is prompted to re-attempt the security questions.
- 6a3. If verification fails 3 times, the claim attempt is locked for 24 hours, logged in the admin audit trail, and the Finder is notified that an unverified claim attempt was made (supports FR-005 moderation).

8a. Duplicate Verified Claim Detected

- 8a1. The system detects the Owner already holds a verified claim linked to this Lost Item report (BR-2).
- 8a2. The system blocks the duplicate verification and shows the existing claim's status with a link to view it.
- 8a3. Use case ends without creating a new claim record.

Exceptions

- System unavailable during verification: the submitted answers are retained locally and the Owner is prompted to retry; no partial claim record is created.
- Notification delivery fails: the claim is still verified and saved; the confirmation is also shown on-screen and stored in the Owner's and Finder's in-app notification history.

Related requirements: FR-001, FR-003, NFR-001 (verification gate), FR-005 (admin moderation of failed/suspicious claims).